

EMS Bike UART Communication Protocol Specification

Version 0.1

1. Purpose

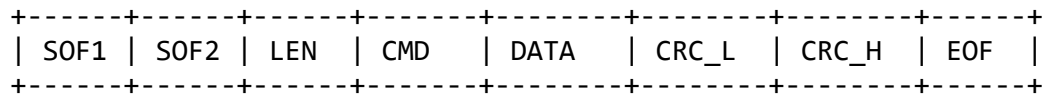
本文件定義 EMS Bike 上控(HMI/APP)與下控(EMS Controller)之 UART 通訊協定。

2. UART Specification

Item	Value
Baud Rate	19200 bps
Data Bits	8
Parity	None
Stop Bits	1
Flow Control	None
Format	8-N-1

3. Packet Format

Frame Structure



SOF

SOF1 =0x55, SOF2 =0xAA

LEN

Definition : LEN = CMD Length + DATA Length

Unit : Byte

Example :

SET_CONTROL

CMD = 1 Byte

DATA = 3 Byte

LEN = 4

CRC16

CRC16-CCITT

Item	Value
Polynomial	0x1021
Initial Value	0xFFFF
Reflect In	No
Reflect Out	No
XOR Out	0x0000

Calculation Range : LEN ~ DATA

Not Included : SOF, CRC, EOF

Transmission Order :

CRC_L

CRC_H

EOF

0x0D

4. Endian

All multi-byte fields :

Little Endian

Example : 0x1234 Transmit 34 12

5. Command Definition

HMI → MCU

CMD	Function
0x01	START
0x02	STOP
0x03	RESET_ERROR
0x06	SHUTDOWN
0x10	SET_CONTROL
0x20	GET_VERSION
0x21	GET_SYSTEM_INFO
0x40	HEARTBEAT

MCU → HMI

CMD	Function
0x80	STATUS_REPORT
0x81	SYSTEM_INFO
0xF0	ACK
0xF1	ERROR_REPORT

6. CONTROL_MODE

```
typedef enum
{
    MODE_KP    = 0,
    MODE_ERG   = 1,
    MODE_DUTY  = 2
}CONTROL_MODE;
```

7. SET_CONTROL

```
typedef struct
{
    uint8_t mode;
    uint16_t target;
}SET_CONTROL_PACKET;
```

MODE_KP : 0 ~ 140

MODE_ERG : Unit : Watt

MODE_DUTY : Unit : 0.1%

Example : 500 = 50.0%

8. ACK_RESULT

```
typedef enum
{
    ACK_OK                = 0,
    ACK_CRC_ERROR         = 1,
    ACK_PARAM_ERROR       = 2,
    ACK_BUSY              = 3,
    ACK_UNKNOWN_CMD       = 4,
    ACK_INVALID_STATE     = 5
}ACK_RESULT;
```

9. STATUS_FLAG

```
typedef enum
{
    STATUS_READY          = 0x01,
    STATUS_RUNNING        = 0x02,
    STATUS_TARGET_REACHED = 0x04,
    STATUS_ERG_LIMIT      = 0x08
}STATUS_FLAG;
```

10. ERROR_FLAG

```
typedef enum
{
    ERR_NONE          = 0x00,
    ERR_COMM_TIMEOUT  = 0x01,
    ERR_RPM_FAIL      = 0x02,
    ERR_POWER_FAIL    = 0x04,
    ERR_COIL_OUTPUT_FAIL = 0x08
}ERROR_FLAG;
```

11. STATUS_PACKET

```
typedef struct
{
    uint16_t rpm;
    uint16_t est_current_mA;
    uint16_t est_watt;
    uint16_t duty_x10;
    uint16_t target_value;
    uint8_t mode;
    uint8_t status;
    uint8_t error;
}STATUS_PACKET;
```

STATUS DATA Length : 13 Bytes

STATUS_REPORT LEN : CMD(1) + DATA(13)= 14= 0x0E

12. HEARTBEAT

Heartbeat Interval : 100 ms

Timeout : 3000 ms

Timeout Action :

EMS OFF

↓

ERR_COMM_TIMEOUT

↓

Wait 5 sec

↓

Power OFF HMI

13. STATUS_REPORT

Report Interval : 100 ms

est_current_mA : Estimated Current

Calculated by:

RPM

+

PWM Duty

+

Current Mapping Table

Note : Not Measured Current

est_watt : Estimated Watt

Calculated by:

RPM

+

Estimated Current

+

EMS Characterization Table

Note : Not Measured Watt

14. Power-On Boot Sequence Example

User Pedaling
↓
RPM > Wake Threshold
↓
MCU Enable HMI Power
↓
HMI Boot Complete
↓
HEARTBEAT
↓
GET_VERSION
↓
VERSION_RESPONSE
↓
GET_SYSTEM_INFO
↓
SYSTEM_INFO_RESPONSE
↓
SET_CONTROL
↓
ACK_OK
↓
START
↓
ACK_OK
↓
STATUS_REPORT
↓
Running

Appendix A - Complete Packet Examples

A-1 SET_CONTROL (KP = 80)

HMI → MCU

55 AA 04 10 00 50 00 12 8D 0D

MCU → HMI

55 AA 03 F0 10 00 3D FF 0D

A-2 SET_CONTROL (ERG = 250W)

HMI → MCU

55 AA 04 10 01 FA 00 97 48 0D

MCU → HMI

55 AA 03 F0 10 00 3D FF 0D

A-3 SET_CONTROL (DUTY = 50.0%)

HMI → MCU

55 AA 04 10 02 F4 01 E9 22 0D

MCU → HMI

55 AA 03 F0 10 00 3D FF 0D

A-4 START

HMI → MCU

55 AA 01 01 1F 3E 0D

MCU → HMI

55 AA 03 F0 01 00 7F CF 0D

A-5 STOP

HMI → MCU

55 AA 01 02 7C 0E 0D

MCU → HMI

55 AA 03 F0 02 00 2C 9A 0D

A-6 RESET_ERROR

HMI → MCU

55 AA 01 03 5D 1E 0D

MCU → HMI

55 AA 03 F0 03 00 1D A9 0D

A-7 SHUTDOWN

HMI → MCU

55 AA 01 06 F8 4E 0D

MCU → HMI

55 AA 03 F0 06 00 E8 56 0D

執行：

EMS OFF

↓

Power OFF HMI

A-8 HEARTBEAT

HMI → MCU

55 AA 01 40 FA 66 0D

Heartbeat 不需 ACK

A-9 GET_VERSION

HMI → MCU

55 AA 01 20 5C 0A 0D

MCU → HMI

55 AA 04 20 01 00 00 74 98 0D

Version : 1.0.0

A-10 GET_SYSTEM_INFO

HMI → MCU

55 AA 01 21 7D 1A 0D

MCU → HMI

55 AA 06 81 8F 0C 01 00 00 40 FE 0D

代表：

VIN = 32.15V

FW = 1.0.0

A-11 ACK_PARAM_ERROR

KP 超出範圍

55 AA 03 F0 10 02 7F DF 0D

A-12 ACK_CRC_ERROR

55 AA 03 F0 10 01 1C EF 0D

A-13 ACK_BUSY

55 AA 03 F0 10 03 5E CF 0D

A-14 ACK_UNKNOWN_CMD

55 AA 03 F0 99 04 B9 1E 0D

A-15 ACK_INVALID_STATE

55 AA 03 F0 01 05 DA 9F 0D

A-16 STATUS_REPORT (ERG Mode)

條件：

RPM	= 90
Estimated Current	= 1500mA
Estimated Watt	= 250W
Duty	= 45.0%
Target	= 250W
Mode	= ERG
Status	= RUNNING + TARGET_REACHED
Error	= NONE

封包：

55 AA
0E
80
5A 00
DC 05
FA 00
C2 01
FA 00
01
06
00
59 C0
0D

單行：55 AA 0E 80 5A 00 DC 05 FA 00 C2 01 FA 00 01 06 00 59 C0 0D

A-17 ERROR_REPORT (POWER_FAIL)

55 AA 02 F1 04 88 C2 0D

A-18 ERROR_REPORT (COMM_TIMEOUT + COIL_OUTPUT_FAIL)

Error：0x09

封包：55 AA 02 F1 09 25 13 0D

Appendix B - CRC Verification Example

Input Data : 04 10 01 FA 00

Expected CRC : 0x4897 Transmit : 97 48

Appendix C - CRC16-CCITT Reference Code

```
#include <stdint.h>

uint16_t CRC16_CCITT(const uint8_t *data, uint16_t len)
{
    uint16_t crc = 0xFFFF;

    while(len--)
    {
        crc ^= ((uint16_t)(*data++) << 8);

        for(uint8_t i = 0; i < 8; i++)
        {
            if(crc & 0x8000)
            {
                crc = (crc << 1) ^ 0x1021;
            }
            else
            {
                crc <<= 1;
            }
        }
    }

    return crc;
}
```

Verification :

```
uint8_t test[] =
{
    0x04,
    0x10,
    0x01,
    0xFA,
    0x00
};

uint16_t crc = CRC16_CCITT(test, 5);
```

```
/* Expected Result = 0x4897 */
```

Appendix D - STATUS_REPORT Field Mapping Table

STATUS_PACKET

```
typedef struct
{
    uint16_t rpm;

    uint16_t est_current_mA;

    uint16_t est_watt;

    uint16_t duty_x10;

    uint16_t target_value;

    uint8_t mode;

    uint8_t status;

    uint8_t error;

}STATUS_PACKET;
```

Field Layout

Offset	Size	Field	Unit
0	2	rpm	RPM
2	2	est_current_mA	mA
4	2	est_watt	W
6	2	duty_x10	0.1%
8	2	target_value	Depends on Mode
10	1	mode	-
11	1	status	Bit Flag
12	1	error	Bit Flag

DATA Length :

13 Bytes

MODE Mapping

Value	Meaning
0	MODE_KP
1	MODE_ERG
2	MODE_DUTY

STATUS Mapping

Value	Meaning
0x01	READY
0x02	RUNNING
0x04	TARGET_REACHED
0x08	ERG_LIMIT

範例：

0x06

RUNNING

+

TARGET_REACHED

ERROR Mapping

Value	Meaning
0x01	COMM_TIMEOUT
0x02	RPM_FAIL
0x04	POWER_FAIL
0x08	COIL_OUTPUT_FAIL

範例：

0x09

COMM_TIMEOUT

+

COIL_OUTPUT_FAIL

Decode Example

封包：

55 AA 0E 80 5A 00 DC 05 FA 00 C2 01 FA 00 01 06 00 59 C0 0D

解析：

Field	Value
RPM	90 RPM
Estimated Current	1500 mA
Estimated Watt	250 W
Duty	45.0 %
Target	250 W
Mode	MODE_ERG
Status	RUNNING + TARGET_REACHED
Error	NONE

CRC：

0xC059